Prof. Stefan Minner
Principal Editor, *International Journal of Production Economics*

Resubmission of a substantially revised manuscript (previously PROECO-D-26-05160)
Replenishing perishables on shared delivery routes: Stock-age-based inventory policies, stochastic lookahead and decision-focused learning

Dear Professor Minner,

Thank you for your decision on PROECO-D-26-05160. You noted that the manuscript did not take a clear perishable inventory management perspective and therefore fell short in comparisons with state-of-the-art methods. I agree with both points. Rather than revising the text, I rebuilt the study around them; the new manuscript shares only the problem domain with the rejected version. I submit it as a new manuscript and summarise the changes below.

1. Perishable inventory management perspective.

- The problem is now a periodic-review perishable inventory system with a routing coupling: every day each policy observes the age-structured stock of every store, decides replenishment, and ships it on shared, capacitated routes; demand is served FIFO, unmet demand is lost and stock past its shelf life is discarded. The rejected version fixed a multi-day plan without observing stock, which is not how perishables are managed.
- Results are reported in inventory terms—cost composition, fill rate, waste rate, delivery frequency and store-level service—and analysed across shelf life, lost-sales/waste cost ratio, fleet tightness, demand variability and issuing policy (FIFO vs. partial LIFO).
- The structural results are stated in inventory terms and corrected: the recourse admits FIFO-optimal solutions once early disposal is allowed (the original lemma was false without it), the marginal value of a delivery is bounded by the lost-sales and waste penalties, and the decision-focused fixed point reduces to the perishable newsvendor fractile. All results are verified numerically.

2. Comparison with state-of-the-art methods.

- The comparator set now contains the perishable inventory rules recommended in the literature—periodic base-stock and base-stock with estimated waste (Broekmeulen and van Donselaar, 2009; Haijema and Minner, 2019)—the deliver-up-to-level rule of Crama et al. (2018) in its state-dependent form, a budgeted robust lookahead (Bertsimas and Thiele, 2006; Solyalı et al., 2012), and the Scenario Predict-then-Optimize method of Jia et al. (2025), together with mean and tuned-quantile lookahead and a perfect-information reference. All policies use the same information and the same forecasts and are tuned on separate validation data.
- Every lookahead decision is obtained from an exact MILP solved to a proven gap of at most 0.5%; the rejected version relied on heuristic fallbacks.
- The results are reported as they are. Across ten settings, stock-age-based base-stock rules with daily delivery are the cheapest policies in nine; only with expensive routing is the scenario-based lookahead of Jia et al. best. Decision-focused targets improve the deterministic lookahead in nine settings and make it the best integrated policy in five, but they do not overturn the advantage of the inventory rules. The managerial conclusions are drawn from this evidence.

Scope and transparency. Demand is generated by a documented contextual model rather than taken from a retailer, which is stated in the abstract and the design section. Earlier claims about M5 calibration, an 81-cell factorial, regret against a heuristic clairvoyant plan and a solver speed-up have been withdrawn. Code, seeds and result files reproducing every number are provided.

I hope that the revised study now meets the scope and standards of IJPE. It has not been published and is not under consideration elsewhere.

Sincerely,

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